

1. 1.2: $d(.5) = .999101$, $d(1) = .996012$, $d(1.5) = .990313$.
1.3: $P = 102.01673$
1.4: Time-0 value of the profit would be 1.01673 per 100 face of 2s of 11/15/11.
2. 2.3 $f(.5) = .25\%$, $f(1) = .4\%$, $f(1.5) = .65\%$
 $\hat{r}(.5) = .25\%$, $\hat{r}(1) = .3250\%$, $\hat{r}(1.5) = .4333\%$
2.5 $P = 100.10016$ The price will be higher in 6 months if the term structure remains unchanged. (Let $r_{.5,T+.5}$ denote the spot rate that will prevail at time .5 for loans initiated at time .5 and settled at time $T + .5$ (semiannual compounding). The statement that the term structure remains unchanged means that $r_{.5,T+.5} = \hat{r}(T)$ for all relative maturities T .)
3. (a) $d(.5) = .96852$, $d(1) = .93351$, $d(1.5) = .89544$
(b) $\hat{r}(1) = .07000$, $f(1) = .07501$
(c) .07475
4. (a) The price of a zero with face value 20,000 and maturity 10 years is 11,654.
(b) The price of an annuity that makes payments of 500 twice a year for the next 10 years is 8,082.5.
(c) The price of a coupon bond with maturity 10 years, face value 7,500 and annual coupon rate .06 is 8,007.375.
5. The customer will have to pay the bank \$107,526.83 at time 2.
6. 3.9248%